

Per-candidate pipeline

Observed spacetime
 $U[t, \mathbf{x}]$



Enumerate candidates
 $(p, \mathbf{s}) \in \mathcal{P} \times \mathcal{S}$



Induce relative-periodic
orbit partition $\{O_j\}$



Project to nearest background
by orbit-wise majority vote



Count orbit statistics
 $n_j, n_j^{(1)}, \hat{\theta}_j$



Score candidate by
 $\text{NML}(p, \mathbf{s}) = \sum_j n_j H_b(\hat{\theta}_j) + \frac{1}{2} \sum_j \log_2 n_j$



Choose minimizer
 $(p^*, \mathbf{s}^*) = \arg \min \text{NML}$

Fast projection
 $O(|U|)$ per candidate

Fit term plus complexity
penalty on orbit classes

Selected periodic
background B^*

Residual mask
 $M = U \oplus B^*$

Residual keeps structured departures
that the scaffold cannot explain

